

Experiment 9

Aim: To demonstrate the performance of Naïve Bayes algorithm by using iris dataset.

Program

```
#naive bayes iris

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()

data = pd.DataFrame(data=iris.data, columns=iris.feature_names)
data['species'] = iris.target

X = data.drop('species', axis=1)
y = data['species']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("accuracy:", accuracy)

conf_matrix = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8, 6))

sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='vanno',
xticklabels=iris.target_names, yticklabels=iris.target_names)

plt.xlabel('Predicted Label')
```

```
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

Output:

